



Automated Artwork **Metadata Generation** using Multimodal LLM

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CHALLENGES

CONTEXT



Cultural institutions are experiencing a significant expansion in the digitization of their archives. Museums and art galleries, in particular, are working to convert extensive collections of artworks into digital formats that can be more easily preserved, searched, and accessed.

STAKEHOLDERS



Museum
Curators

Manual annotation is:

- time-consuming;
- expensive;
- non-scalable.



AI Researchers

- Limited availability of high-quality labeled datasets;
- Lack of standardized metadata schemas across institutions.

CHALLENGES

Reduce the time and effort required for manual metadata compilation

Overcome the scarcity of reliable labeled data and the inconsistency of metadata structures across different cultural institutions

OBJECTIVES



Allowing curators to focus on **higher-value activities** through the automatic generation of metadata.



Enriching a database with more detailed information **improves searchability**, facilitates research, and enhances the understanding of the content.



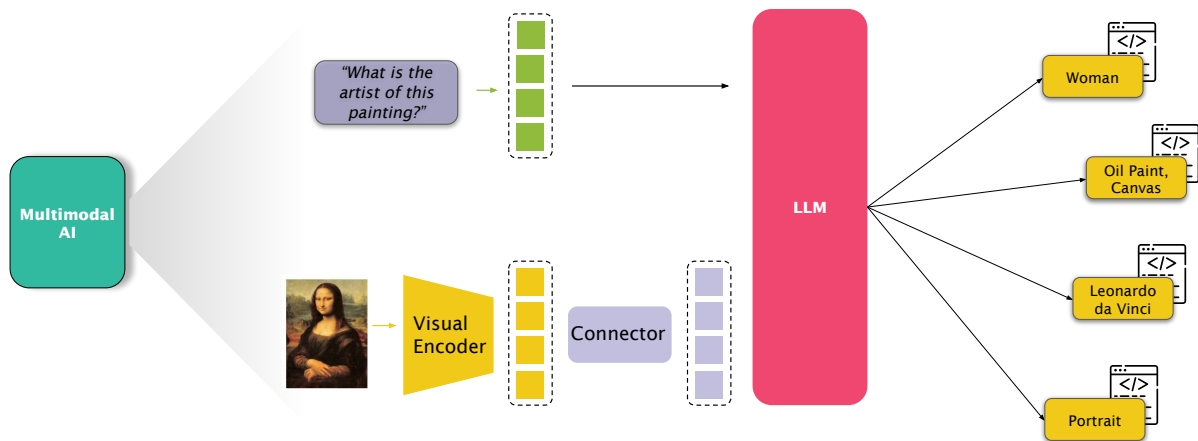
Standardize data structures to enable easier **information sharing** and improve public access to cultural heritage.



HYPOTHESES

Q: Can a multimodal LLM correctly provide informations about a painting?

A: MLLM can effectively bridge the gap between raw visual data and structured knowledge



METHODOLOGY



ZERO-SHOT EVALUATION



Evaluating the models **"As-Is"** (Pre-trained weights) to benchmark their intrinsic capabilities without initial fine-tuning.

- **PROMPT ENGINEERING**

To mitigate bias and hallucinations, we designed a **"Stringent Prompt Template"**. This constrains the model to output objective metadata rather than free-form creative text.

- **VLM SELECTION**

We used **BLIP-2** and **Qwen2-VL** as our core Vision-Language Models to test varying architectures (Q-Former vs. Native Resolution).



FINE-TUNING

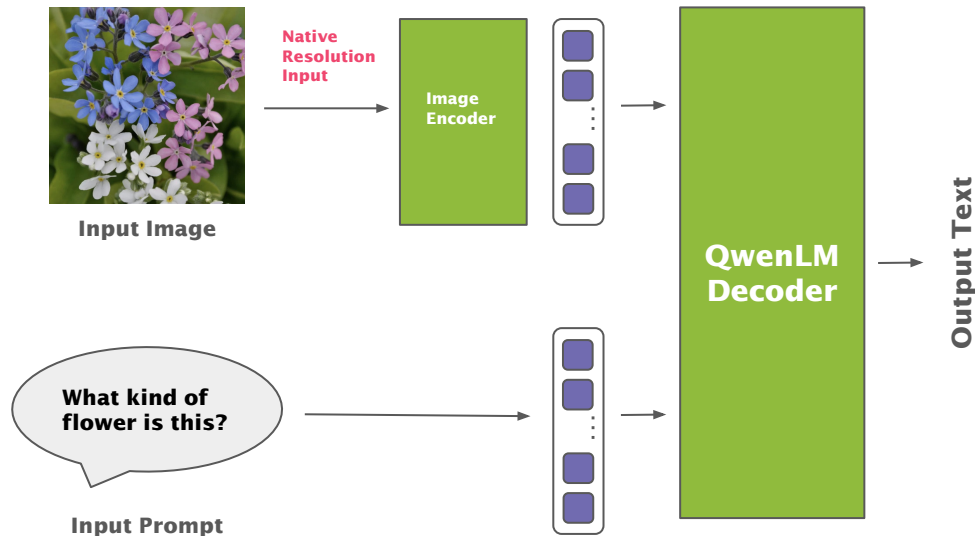


Fine-tune the model that demonstrates the highest baseline performance in zero-shot benchmarks.

- **PEFT TECHNIQUES**

Updates only a small subset of parameters avoiding to retrain the entire model.

QWEN2-VL-2B-INS.



CHAT-TEMPLATE

Allow complex system prompts, enabling the model to handle long contexts and advanced reasoning.

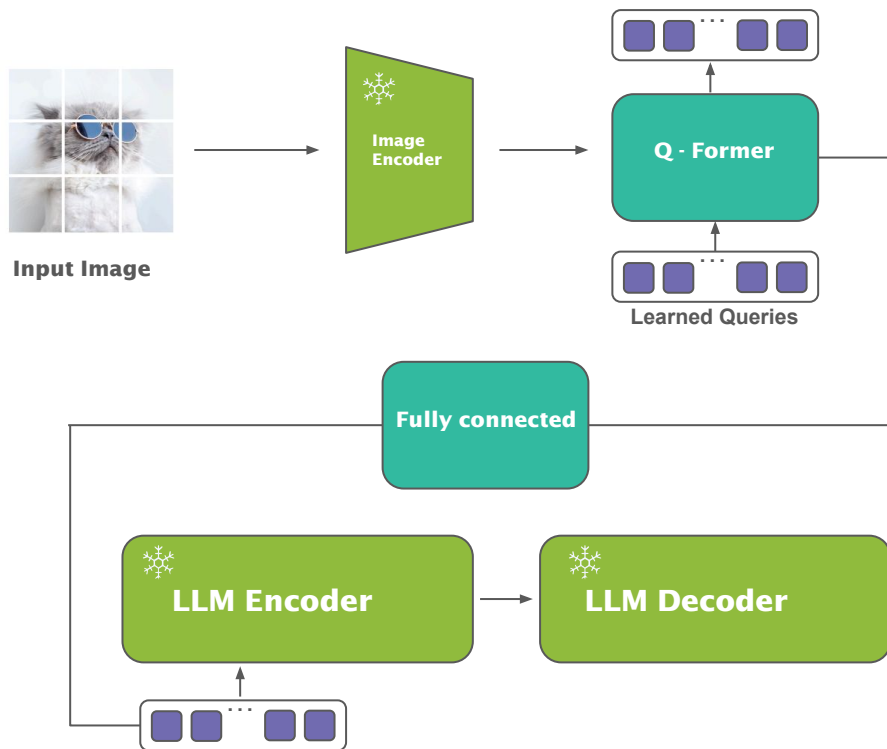
NATIVE RESOLUTION

Processes images at their native resolution without fixed-size compression, ensuring maximum clarity for fine details.

MULTIMODAL ROTARY POSITION EMBEDDING

Decomposes position into time, height, and width. Captures 3D geometry within linear 1D sequences.

BLIP-2 FLAN-T5-XL



INSTRUCTION-TUNED

What are shown in the photo?



A man and a chicken.

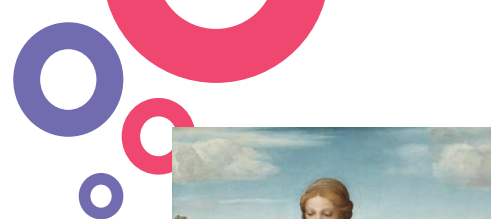
FROZEN

During pre-training Image Decoder and LLM are frozen to avoid catastrophic forgetting

INFORMATION BOTTLENECK

The Q-Former acts as an information bottleneck between the frozen image encoder and the frozen LLM

DATASET

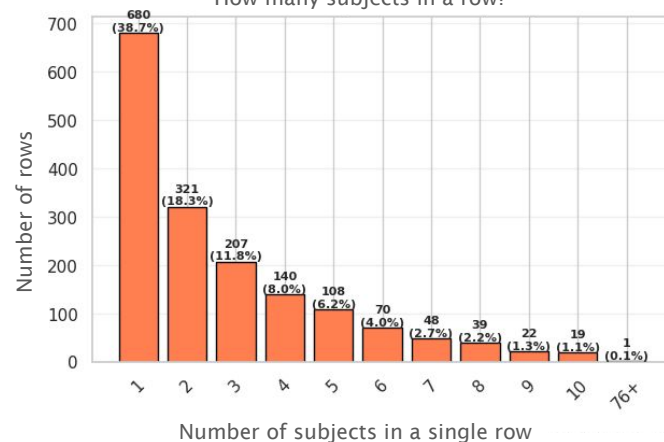


artefact	type	title	description	genres	materials	subjects	authors	images
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https://wikidata	Painting	Madonna del Prato	painting by Raphael	['religious art']	['oil paint', 'poplar panel']	['woman', 'boy', 'Mary', 'Christ Child',...]	['Raphael']	02992163-d6..
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Column	Unique elements	Total occurrences	Average elements/row	Max elements/row
Authors	1.040	1.773	1,01	2
Genres	41	1.882	1,07	4
Materials	47	3.494	1,99	5
Subjects	2.442	6.446	3.67	76

How many subjects in a row?



EXPERIMENTS

HYBRID TASK

- **QA problem:** VLM generate an answer as close as possible to the expected Wikidata answer.
- **Open-domain vocabulary:** the possible classes are not defined in advance.

SYNTACTIC EVALUATION

How many terms are correctly predicted? We cannot use standard precision, recall, F1 → **EXACT MATCH**

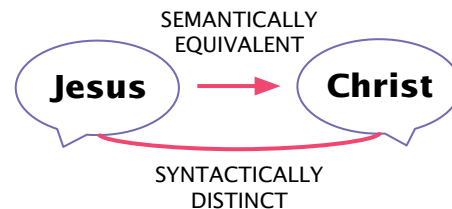
SEMANTIC EVALUATION

Exact Match cannot capture semantic similarity → **MPNET EMBEDDINGS**

⚠ Similarity('person', any author) ≥ 0.7

QA METRICS

- **BLEU** and **ROUGE** useful for the N-gram overlap analysis.
- These metrics expect answers with more than one token. Particularly, BLEU penalizes answer with only one token.



CLASSES

OUR



EXACT MATCH EXTENSION

Querying Wikidata database to retrieve aliases of authors



WIKIDATA

QUANTITATIVE ASSESSMENT

QWEN

"" You are an expert art historian. Analyze this artwork and identify the genre depicted. Provide structured JSON with genres attribution.

Steps:

1. Visual Analysis: Describe style, technique, color palette, composition, subject matter, and period characteristics
2. Genre Indicators: Identify distinctive features (subject matter, composition conventions, thematic elements, stylistic patterns) that indicate specific genre(s)
3. Mixed Genre Detection: Note any characteristics suggesting hybrid or multiple genre influences
4. Classification: Determine the most likely genre(s) based on visual evidence

Rules: Identify ANY genre(s) from visual analysis. List ALL genre(s).

OUTPUT ONLY valid JSON:

```
{ "genres": [ "genre1", "genre2", "genre3" ] }
```

CRITICAL: Your response must be valid JSON. Always close with } ""

BLIP

""

What are the artistic genres of this painting?

Genre:

""

	Exact Match		Similarity Score		BLEU		ROUGE-1	
	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP	QWEN	BLIP
Authors	0.150	0.001	0.800	0.684	X	X	0.154	0.003
Subjects	0.014	0.000	0.717	0.722	0.030	0.014	0.193	0.099
Materials	0.218	0.000	0.886	0.838	0.210	0.105	0.609	0.478
Genres	0.064	0.353	0.586	0.549	0.079	0.077	0.324	0.485

QUALITATIVE ASSESSMENT

Strength

Weakness

Authors

—

Inconsistent
output
normalization

Materials

High accuracy on
standard "Oil on
Canvas"

Texture
Recognition

Genres

Macro-
classification

Contextual
Nuance

Subjects

Comprehensive
Visual
Description

Iconographic
Hallucination

Authors

Prediction	Ground truth	Correctness
Giovanni Antonio Canal	Canaletto	✓
Michelangelo Buonarroti	Buonarroti Michelangelo	✓
Thomas Moran	Frederic Edwin Church	✗

Materials

Prediction	Ground truth	Correctness
Oil on canvas	oil paint	✓
Wood, paint	oak wood	✓
Oil on canvas	poplar panel	✗

Genres

Prediction	Ground truth	Correctness
landscape, landscape with windmill	landscape painting	✓
landscape, cityscape, nautical	Veduta	✓
landscape, cityscape, nautical	Historical mythological painting	✗

Subjects

Prediction	Ground truth	Correctness
man in armor, sword	Charles X Gustav of Sweden	✓
lemons, carafe, tablecloth	fruit	✓
King George I	William III of England	✗

CONCLUSION



OBJECTIVE

The project aims to streamline the management of cultural heritage data by automating metadata generation, enriching databases with detailed information, and standardizing data structures to improve accessibility, searchability, and knowledge sharing.

METHOD

The method involves using an AI system based on a pre-trained large language and vision model (Qwen), which is fine-tuned to automatically generate metadata from images.

This is **effective to reach the objective** because the method leverages advanced AI capabilities to interpret complex visual content and generate accurate metadata automatically, reducing manual effort and ensuring consistency.

NEXT-STEP

Fine-tuning of Qwen applying PEFT technique.

THANK YOU

USER REQUIREMENTS

ID Req	Requirement description	Status	Type of requirement	Priority
1	Accept an image and a text query, with the option to specify an expected metadata/JSON schema	In progress	Functional	Must have
2	Support efficient adaptation of the MLLM through parameter-efficient fine-tuning techniques, such as Low-Rank Adaptation (LoRA) or adapters	To do	Functional	Must have
3	Accuracy: the system must achieve minimum (F1, Recall, Precision) thresholds in critical fields (e.g. subjects, authors)	In progress	Non-Functional	Must have
4	Extraction of metadata producing structured and machine-readable output	In progress	Functional	Must have
5	Robustness: performance must degrade in a controlled manner as image quality, resolution, lighting and noise vary, avoiding gross errors or hallucinations.	In progress	Non-Functional	Must have
6	Allow expert to provide feedback on generated metadata	Future	Functional	Could to have